

August 6, 2026

Dear Hiring Team,

I am writing to apply for the Software Developer position at Promise Robotics. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, C++, machine learning, algorithmic software, backend APIs, testing-minded development, and research experimentation, I am excited by the opportunity to help build software systems for AI-powered robotic manufacturing.

My graduate research focused on building scalable Python and C++ systems for complex Earth observation data. I designed DGGS-based algorithms and encoding pipelines, automated data ingestion and preprocessing, implemented structured tensor storage and retrieval workflows, and improved processing performance with C++ multithreading. This work required the same engineering instincts Promise Robotics emphasizes: careful software design, performance analysis, experimentation, debugging, and turning research ideas into robust working systems.

I also bring applied machine learning experience that aligns with the role's AI and advanced-algorithm focus. In my DEM super-resolution project, I trained and evaluated PyTorch ResNet models, built data collection and preprocessing workflows, and assessed reconstruction quality with metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. As a Deep Learning Teaching Assistant with Neuromatch Academy, I supported students through hands-on PyTorch tutorials covering CNNs, transformers, diffusion models, and reinforcement learning, which strengthened both my technical foundation and my ability to communicate ML concepts clearly.

Before graduate school, I worked as a backend developer building Django REST APIs for NLP, text-to-speech, and speech-to-text services backed by PostgreSQL. I implemented authentication, payment-service logic, and service-plan workflows, and contributed to a chatbot system using GPT-style responses and FAQ data. This experience gives me a practical software-development base beyond research code, including API design, database-backed applications, and user-facing reliability.

Promise Robotics is especially interesting to me because it sits at the intersection of AI, robotics, manufacturing, and construction, with software directly tied to real physical outcomes. I do not claim professional robotics-manipulator experience yet, but I bring strong Python, ML, algorithms, C++/systems thinking, software experimentation, communication, and Calgary availability. I would be excited to learn the robotics and manufacturing domain deeply while contributing immediately to software algorithms, testing, and cross-functional development.

Thank you for considering my application. I would welcome the opportunity to discuss how my Python, machine learning, algorithmic software, backend, and research experimentation experience can contribute to Promise Robotics' core manufacturing technologies.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com